

XClipGS: Exact Half-Space Clipping for Medical Volume Gaussian Splatting

Zhongpai Gao¹, Benjamin Planche¹, Meng Zheng¹, Anwesa Choudhuri¹, Chaoyi Zhou²,
Terrence Chen¹, Ziyang Wu¹

¹United Imaging Intelligence, Boston MA, USA ²Clemson University, Clemson SC, USA
{first.last}@uii-ai.com



Figure 1: **Faithful interactive clipping.** XClipGS, reference volume rendering, and ClipGS (Li et al. 2025a) (left to right) on a held-out test cut. The arrow marks ClipGS leakage; labels report quality and speed.

Abstract

Gaussian-splatting proxies enable interactive rendering of volumetric medical scans, but a clipping plane exposes anatomy not constrained by external-view training and intersects primitives that conventional splatting can only keep or drop whole. We present XClipGS (*eXact Clipping*), which treats these as two separate problems: the render-time clip operator and supervision of the hidden interior. Under the local affine model used by EWA splatting, each half-space-restricted Gaussian’s ray integral factorizes exactly into its ordinary 2D footprint and a conditional Gaussian CDF whose argument is affine in pixel coordinates. The resulting closed-form per-splat operator introduces no learned clipping parameters or auxiliary network and remains differentiable with respect to the primitive and plane. We use multi-distance reference views with varied clipping-plane axes and offsets to supervise the interior through the same operator. We also introduce a paired clipped/unclipped cut-face protocol with difference-referenced cut error (CDE) and culled-side leakage (Leak), because global image metrics dilute errors near the plane. On eight CT and MRI volumes with plane offsets not used for training, XClipGS attains the highest PSNR on every volume (33.56 versus 32.34 dB for ClipGS) while rendering at over 650 FPS, far above real time, versus 278 FPS. On voxel-axis cut-face views it raises average band SSIM from .809 to .860 and leaks roughly $41\times$ less. Without retraining, it also achieves the best average across all four metrics on arbitrary-normal planes; on a fixed interior it matches RaRa’s

face fidelity with about $16\times$ less leakage.

1 Introduction

CT and MRI are inspected by orbiting, zooming, and clipping the volume to expose hidden anatomy (Weiskopf, Engel, and Ertl 2003; Niedermayr et al. 2024). Cinematic volume rendering makes these cross-sections legible through realistic depth and material cues, but converged rendering remains expensive on commodity and web hardware. Gaussian splatting (Kerbl et al. 2023) offers a real-time proxy: a volume can be represented by compact primitives and rasterized at interactive rates. Existing medical proxies are usually trained from views surrounding the intact volume, however, so their supervision concentrates on the visible outer anatomy.

An interactive clipping plane changes the rendering problem. It removes the outer layers and turns a previously hidden cross-section into the dominant surface. A proxy that matches external views can therefore fail abruptly when a user moves the plane through its interior. This challenges representation generalization and requires the renderer to determine each intersected Gaussian’s partial contribution.

We separate these effects into two failure modes. The **clip operator** must truncate primitives intersected by the plane: binary keep/drop schemes, including ClipGS’s trainable decision (Li et al. 2025a), quantize the boundary, while RaRa’s ray–ellipsoid chord ratio (Li et al. 2025b) only approximates

partial contribution. The **unsupervised interior** is equally important: an accurate operator reveals whatever anatomy the proxy learned behind the outer surface. External views can leave that anatomy ambiguous, while supervision through an approximate operator can teach the proxy the operator’s boundary errors. Faithful clipping therefore requires both an accurate partial-primitive rule and training views that reveal the hidden volume: with the exact operator in place, removing the clipped training views still costs 8.3 dB on held-out clipped views across all eight volumes.

XClipGS (*eXact Clipping*) addresses both with a closed-form, differentiable per-pixel clip and clip-aware interior supervision (Figure 1). Under the local affine model of EWA splatting (Zwicker et al. 2001), each half-space-restricted Gaussian’s ray integral equals its ordinary 2D footprint times a conditional Gaussian CDF whose argument is affine in pixel position. The clip is exact under affine EWA at the per-splat integration level and requires no learned clipping parameters or auxiliary network. Rendering the proxy through this operator against clipped reference views supplies direct gradients to primitives near the exposed face. We vary camera distance, plane axis, and offset so that training includes full-volume context and detailed, diverse cross-sections. The construction acts on primitive geometry and applies unchanged to CT and MRI within the same training and rendering pipeline.

Our experiments distinguish whole-system quality from operator quality. We compare the analytic operator with a hard cull (HC) and the forward-KL-optimal moment-matched Gaussian surrogate (MM) under a shared N-DGS backbone (Gao et al. 2025a). A 3DGS-based ClipGS (Li et al. 2025a) provides a system comparison, while operator swaps on fixed checkpoints isolate the render-time rule and include RaRa (Li et al. 2025b). Numerically held-out plane offsets test interpolation along the sampled voxel axes, while a separate test uses arbitrary plane orientations without retraining. Paired clipped and unclipped reference renders localize error at the cut face, where global metrics are least informative.

In summary, this paper contributes:

- **Analytic half-space clipping:** a closed-form, differentiable per-pixel Gaussian integral that is exact for each primitive under the affine EWA model and introduces no learned clipping parameters or auxiliary network;
- **Clip-aware interior supervision:** multi-distance reference views whose clipping-plane axes and offsets vary across the volume, with evaluation on held-out offsets and arbitrary orientations absent from training, testing transfer to new cross-sections without retraining;
- **A CT and MRI benchmark with cut-face metrics:** eight volumes with held-out planes and metrics that score the cut where global image quality is nearly blind to it, on which our method achieves the best average on all four metrics for both plane sets. Parameter sweeps verify that localization and leakage conclusions are stable across thresholds, band widths, and leakage-mask settings.

2 Related Work

Medical volume rendering and Gaussian proxies. Cinematic medical volume rendering path-traces global il-

lumination through the volume (Kroes, Post, and Botha 2012; Comaniciu et al. 2016; Novák et al. 2018), making cross-sections legible through realistic depth and material cues (Dappa et al. 2016). Converged rendering is not interactive, progressive estimates can flicker (Martschinke et al. 2019), and web deployment remains difficult (Xu et al. 2023). Niedermayr et al. (2024) reproduce cinematic anatomy in real time with a Gaussian proxy, but its clip planes are fixed during preprocessing. Other Gaussian methods target rendering, reconstruction, or novel-view synthesis (Gao et al. 2024; Cai et al. 2024; Zha et al. 2024; Yang et al. 2024; Peng et al. 2025; Yang et al. 2026), not partial-primitive clipping.

Geometric clipping of Gaussian splats. ClipGS (Li et al. 2025a), our primary medical baseline, makes a binary per-primitive decision trainable through an STE and deforms primitives with a plane-conditioned MLP. RaRa Clipper (Li et al. 2025b) scales intersecting primitives by the visible ray–ellipsoid chord fraction, and its released formulation has no backward pass for clip-aware training. Earlier medical work toggles opacity layers (Kleinbeck et al. 2025). Our operator instead integrates the half-space-restricted Gaussian ray density under affine EWA and supports both supervision and rendering. Section 4 compares the system with ClipGS and isolates RaRa as a render-time rule on a fixed interior.

Conditioned Gaussian splatting. 6DGS (Gao et al. 2025a) and 7DGS (Gao et al. 2025b) form N-dimensional Gaussian splatting (N-DGS), with Beta-conditioned generalizations (Liu et al. 2026, 2025). Our opacity-conditioned N-DGS backbone leaves each primitive an ordinary 3D Gaussian, so geometric clipping applies unchanged; its statistical “slicing” is distinct from a planar cut. Render-FM (Gao et al. 2026) supplies initialization.

Gaussian-splatting fidelity. Our derivation adopts the local affine EWA projection used by 3DGS (Kerbl et al. 2023; Zwicker et al. 2001). Exact perspective-ray methods such as 3DGEER (Huang et al. 2026) change the projection model and require re-derived coefficients. Our exactness claim accordingly concerns the per-primitive affine-EWA ray integral; the base renderer’s splat order and alpha compositing remain unchanged. Analytic-Splatting integrates over projected pixels, while Mip-NeRF and Mip-Splatting address pixel-footprint sampling (Liang et al. 2024; Barron et al. 2021; Yu et al. 2024); these concerns are complementary to half-space restriction.

Semantic editing and half-space representations. Semantic editors select or remove whole primitive subsets (Chen et al. 2024; Ye et al. 2024; Cen et al. 2025; Zhou et al. 2024; Jain, Mirzaei, and Gilitschenski 2024). 3D-HGS instead learns a fixed half-Gaussian split (Li et al. 2025c); neither evaluates the continuous contribution of a user-supplied plane. EVSplitting replaces both sides with moment-preserving Gaussians (Feng et al. 2024). Like our truncated-normal MM ablation (Tallis 1961), it re-Gaussianizes; our operator integrates the restricted density.

3 Method

Overview. XClipGS combines an exact per-splat render-time clipping operator with clip-aware supervision that learns

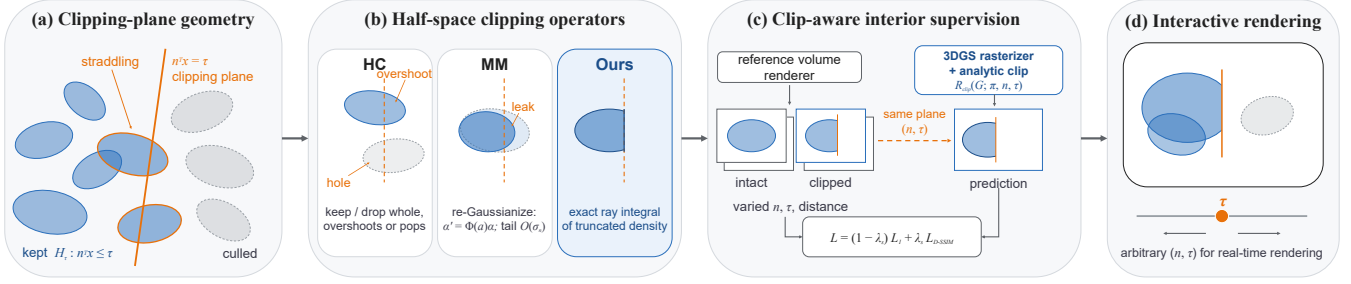


Figure 2: **XClipGS pipeline.** Exact per-splat half-space integration clips straddling Gaussians, while multi-distance cut views supervise the hidden interior through the same differentiable operator.

the hidden interior (Figure 2). The operator analytically integrates plane-straddling Gaussians under affine EWA, while multi-distance clipped views with varied plane axes and offsets train the interior through the same differentiable rule. We first establish the rendering model, derive the analytic operator and its hard-cull and moment-matched reference points, and then describe interior supervision.

3.1 Setup and notation

A scene is a set of Gaussian primitives $\{(\mu_i, \Sigma_i, \alpha_i, \mathbf{c}_i)\}_{i=1}^N$ with mean $\mu_i \in \mathbb{R}^3$, covariance $\Sigma_i \in \mathbb{R}^{3 \times 3}$, opacity $\alpha_i \in [0, 1]$, and view-dependent color $\mathbf{c}_i$ (Kerbl et al. 2023). Following the EWA formulation (Zwicker et al. 2001), a camera induces a local affine approximation of the projection at each primitive center,

$$\mathbf{u}(\mathbf{x}) \approx \hat{\mathbf{p}}_i + \mathbf{M}_i (\mathbf{x} - \mu_i), \quad \mathbf{M}_i = \mathbf{J}_i \mathbf{W} \in \mathbb{R}^{2 \times 3}, \quad (1)$$

where $\mathbf{W}$ is the world-to-camera rotation, $\mathbf{J}_i$ the projection Jacobian at μ_i , and $\hat{\mathbf{p}}_i$ the projected center. The screen-space covariance is $\mathbf{A}_i = \mathbf{M}_i \Sigma_i \mathbf{M}_i^\top$, and the contribution of primitive i to a pixel $\mathbf{p}$ is $\alpha_i \mathcal{K}_G(\delta_i; \mathbf{A}_i)$ with $\delta_i = \hat{\mathbf{p}}_i - \mathbf{p}$ and $\mathcal{K}_G(\delta; \mathbf{A}) = \exp(-\frac{1}{2} \delta^\top \mathbf{A}^{-1} \delta)$; contributions are alpha-composited front-to-back. This per-pixel weight is, up to normalization, the integral of the 3D Gaussian density along the pixel’s viewing ray under the affine approximation (1), an identity we exploit below. Our operator changes this per-primitive weight but retains the base rasterizer’s depth order and front-to-back compositing.

Clipping plane. A clipping plane with unit normal $\mathbf{n}$ and offset τ keeps the half-space $\mathcal{H}_\tau = \{\mathbf{x} \in \mathbb{R}^3 : \mathbf{n}^\top \mathbf{x} \leq \tau\}$. The reference renderer removes density outside $\mathcal{H}_\tau$ before ray integration through its volumetric half-space clipping interface. XClipGS and all operator ablations accept arbitrary $(\mathbf{n}, \tau)$. For obliquely acquired scans, voxel-axis planes are generally rotated and noncanonical in world coordinates; the arbitrary-normal evaluation instead specifies planes directly in physical-world coordinates. Throughout, let

$$\sigma_n^2 = \mathbf{n}^\top \Sigma \mathbf{n}, \quad a = \frac{\tau - \mathbf{n}^\top \mu}{\sigma_n}, \quad \lambda(a) = \frac{\phi(a)}{\Phi(a)}, \quad (2)$$

denote the primitive’s variance along the plane normal, its standardized signed distance to the plane, and the inverse Mills ratio, where ϕ and Φ are the standard normal density and CDF. $\Phi(a)$ is precisely the fraction of the primitive’s mass on the kept side.

3.2 Half-space clipping operators

Clipping operators. A clip operator specifies how a primitive contributes when rendering under $\mathcal{H}_\tau$. The target primitive is the unnormalized restricted density $r(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \mu, \Sigma) \mathbb{1}[\mathbf{x} \in \mathcal{H}_\tau]$; our operator evaluates its ray integral in closed form (Proposition 1). We contrast it with two reference points. A hard cull represents the whole-primitive decisions used by prior work (Li et al. 2025a); a moment-matched truncation is a smooth Gaussian approximation derived from classical truncated-normal statistics. Comparing them in one renderer isolates the cost of binary truncation and re-Gaussianization relative to integrating the restricted density itself. We abbreviate the hard cull, moment-matched truncation, and our analytic operator as HC, MM, and Ours.

Hard cull (HC). This operator keeps or drops each primitive *whole* using $\mathbb{1}[\mathbf{n}^\top \mu \leq \tau]$. It adds no per-pixel factor, but straddling primitives overshoot or vanish, producing a center-quantized boundary as the plane sweeps. The raw decision is also non-differentiable. ClipGS trains its per-primitive decision through a straight-through estimator and deforms primitives; HC isolates the underlying binary operator from those learned components.

Moment-matched truncation (MM). This surrogate, which we derive from classical statistics rather than take from prior work, replaces each straddling primitive by the Gaussian whose first two moments match the normalized truncated density $r/\Phi(a)$. Equivalently, it minimizes $D_{\text{KL}}(r/\Phi(a) \parallel q)$ over Gaussian q . The required one-sided truncated-normal moments (Tallis 1961), propagated through the joint covariance, use $\mathbf{d}_n = \Sigma \mathbf{n} / \sigma_n$ and are

$$\begin{aligned} \alpha' &= \Phi(a) \alpha, & \mu' &= \mu - \lambda(a) \mathbf{d}_n, \\ \Sigma' &= \Sigma + [v(a) - 1] \mathbf{d}_n \mathbf{d}_n^\top. \end{aligned} \quad (3)$$

where $v(a) = 1 - a \lambda(a) - \lambda(a)^2$. Opacity scales by the kept mass, the mean shifts to the truncated centroid, and the variance contracts along the clip normal by the truncated-variance factor $v(a) \in (0, 1]$. MM is smooth in $(\mu, \Sigma, \alpha, \tau)$ and can supervise the cut face directly. Its limitation is intrinsic: re-Gaussianization restores unbounded support, so some mass still renders beyond the plane and the cut becomes a smooth tail of width $\mathcal{O}(\sigma_n)$ rather than an edge.

Exact per-pixel integral (Ours). We render the truncated density itself rather than re-Gaussianizing it. Under the affine

approximation (1), screen position $\mathbf{u}$ and clip coordinate $\omega = \mathbf{n}^\top \mathbf{x}$ are jointly Gaussian. Conditioning on $\mathbf{u}=\mathbf{p}$ factorizes the truncated ray integral into the unclipped footprint and $P(\omega \leq \tau \mid \mathbf{u}=\mathbf{p})$. With $\mathbf{b} = \mathbf{M}\Sigma\mathbf{n} \in \mathbb{R}^2$, the conditional variance is

$$s^2 = \sigma_n^2 - \mathbf{b}^\top \mathbf{A}^{-1} \mathbf{b}, \quad (4)$$

and the clipped per-pixel contribution becomes

$$\boxed{\begin{aligned} \alpha \mathcal{K}_G(\boldsymbol{\delta}; \mathbf{A}) \cdot \Phi(\ell(\boldsymbol{\delta})), \quad \ell(\boldsymbol{\delta}) &= \mathbf{k}^\top \boldsymbol{\delta} + h, \\ \mathbf{k} &= \frac{\mathbf{A}^{-1} \mathbf{b}}{s}, \quad h = \frac{\tau - \mathbf{n}^\top \boldsymbol{\mu}}{s}. \end{aligned}} \quad (5)$$

Up to normalization, this Gaussian–CDF product has the classical selection-normal form (Arellano-Valle, Branco, and Genton 2006). We specialize it to half-space EWA ray integration and derive the rasterizer coefficients and gradients. The clip thus reduces to three transient coefficients $(\mathbf{k}, h)$ computed from existing geometry per primitive and view, plus one CDF evaluation per sample with argument affine in the pixel offset $\boldsymbol{\delta}$ already available to the rasterizer. These are render-time intermediates, not learned parameters.

Why the factor is per pixel. The marginal kept-mass fraction $\Phi(a)$ from Equation 2 is a single scalar for the whole primitive. Multiplying the projected footprint uniformly by this value would attenuate the splat but could not locate the cut within it. Equation 5 instead uses the conditional probability given the pixel. The term $\mathbf{k}^\top \boldsymbol{\delta}$ moves the transition across the footprint, while h shifts it as the plane moves. Consequently, pixels whose rays pass through the retained side approach a factor of one and those on the removed side approach zero, with the transition width determined by the remaining uncertainty s along the ray.

Proposition 1 (Exactness). *Under the local affine approximation (1), the contribution (5) equals the viewing-ray integral of the unnormalized half-space-restricted density r , up to the global normalization shared with the unclipped splat.*

Conditioning the jointly Gaussian screen and clip coordinates on $\mathbf{u}=\mathbf{p}$ factorizes the ray integral into the EWA footprint and $P(\omega \leq \tau \mid \mathbf{u}=\mathbf{p})$. Its Schur-complement variance is Equation 4, and its affine conditional mean yields Equation 5. Supplement S1 gives the full derivation.

Scope and boundary behavior. Proposition 1 is exact for one primitive’s ray integral within the affine EWA model (1) (Zwicker et al. 2001) and adds no approximation to its unclipped footprint. At image level, XClipGS retains the base rasterizer’s standard center-based splat order and front-to-back alpha compositing; the proposition isolates the added half-space integral from these established renderer choices. A perspective-ray formulation requires re-derived conditional coefficients (Huang et al. 2026). Within affine EWA, the boundary width is $s \leq \sigma_n$; as the plane becomes parallel to the rays, $s \rightarrow 0$ and $\Phi(\ell)$ approaches a per-pixel step without MM’s tail.

Differentiability. For $s > 0$ our operator is smooth in primitive parameters and in τ ; at the degenerate $s \rightarrow 0$ limit the factor becomes a step (a plane parallel to the ray). Since

$\partial_\ell \Phi = \phi(\ell)$, gradients pass through $(\mathbf{k}, h)$ and the center to $(\boldsymbol{\mu}, \Sigma)$. Thus MM and Ours supervise the cut itself. HC has no gradient through its binary decision and updates only kept primitives, the discontinuity ClipGS addresses with an STE.

3.3 Clip-aware interior supervision

Unconstrained interiors. A proxy trained only from external views leaves hidden density weakly constrained: primitives behind the visible surface receive gradients only through accumulated transmittance, and the anatomy exposed by a new cut was never observed directly. Interior fidelity therefore depends on supervision as well as the clip operator. The same external images can be explained by different interior configurations that disagree once the front layers are removed. Clipped views resolve this ambiguity by directly exposing hidden density, not by adding cameras around the intact outer surface alone. The analytic operator makes the cut itself differentiable, turning interior supervision into ordinary photometric training: gradients from clipped reference views flow through Φ to primitives at the exposed face.

Training-view generation. For each scan, the reference volume renderer generates both intact and clipped views across a range of camera distances. Intact views orbit and frame the whole scan. For clipped views, we sample one of the three voxel axes and an offset $\tau \sim \mathcal{U}[\tau_{\min}, \tau_{\max}]$ spanning the volume core, then aim the camera obliquely at the exposed face. Each frame records its plane $(\mathbf{n}, \tau)$, and the same plane is passed to the rasterizer during training. Varying distance supplies both full-volume context and close-up detail; varying axis and offset exposes diverse cross-sections rather than a single prescribed cut.

Training objective. Training minimizes the standard 3DGS photometric loss through the differentiable clip (Kerbl et al. 2023; Wang et al. 2004),

$$\mathcal{L} = (1 - \lambda_s) \mathcal{L}_1(\mathcal{R}_{\text{clip}}(\mathcal{G}; \pi_v, \mathbf{n}_v, \tau_v), I_v) + \lambda_s \mathcal{L}_{\text{D-SSIM}}, \quad (6)$$

where $\mathcal{R}_{\text{clip}}$ renders the primitive set $\mathcal{G}$ under camera π_v at plane $(\mathbf{n}_v, \tau_v)$ ($\tau_v = \infty$ for intact views), I_v is the reference render, and $\lambda_s = 0.2$ in all experiments. Gradients from the exposed cross-section flow through Φ into the position, covariance, and opacity of interior primitives near the cut. Section 4 tests both held-out offsets along the training axes and arbitrary plane orientations without retraining.

3.4 Implementation

Rasterization. Our CUDA implementation combines Speedy-Splat culling (Hanson et al. 2025), TC-GS (Liao et al. 2025), and Mip-Splatting’s 3D filter (Yu et al. 2024). Preprocessing computes the shared truncation statistics. MM then rewrites each straddling primitive once, whereas Ours retains its ordinary EWA footprint and evaluates $\Phi(\ell)$ only for samples from straddling primitives. Both paths remain differentiable end to end and accept arbitrary plane orientations. Supplements S2–S3 give numerical safeguards, gradient validation, and experimental configurations.

Table 1: **Held-out quality and speed on eight volumes.** FPS is measured at 1600×1600 on an A100. **Bold** marks the best.

Scene	XClipGS (Ours)				ClipGS (reimpl.) (2025a)				MM				HC			
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FPS $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FPS $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FPS $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FPS $\uparrow$
abdomen	33.30	.965	.083	682	32.20	.960	.089	293	32.97	.964	.084	615	33.02	.963	.085	651
intestine	31.77	.949	.093	597	30.83	.944	.100	220	31.49	.948	.094	575	31.50	.947	.095	599
knee-joint	29.14	.913	.185	586	27.89	.904	.197	252	28.85	.912	.186	580	28.44	.908	.191	604
lower-limb	34.28	.980	.026	670	33.02	.977	.028	241	34.20	.980	.026	653	34.04	.980	.026	693
vascular	36.38	.975	.031	666	34.68	.971	.035	320	36.18	.975	.031	621	36.10	.974	.032	654
heart	29.20	.940	.095	603	28.14	.930	.106	239	28.72	.938	.097	579	28.65	.936	.101	594
nose (MRI)	35.92	.977	.047	699	34.65	.973	.051	306	35.67	.976	.048	698	35.55	.975	.049	694
hand (MRI)	38.48	.978	.036	740	37.30	.974	.041	353	38.22	.977	.037	724	37.85	.975	.040	772
avg	33.56	.960	.075	655	32.34	.954	.081	278	33.29	.959	.075	631	33.14	.957	.077	658

4 Experiments

We evaluate held-out plane offsets and orientations, cut-face errors, and render-time operator swaps. Ours, MM, and HC share the N-DGS backbone, initialization, views, and planes; ClipGS uses the same initialization, views, planes, and iteration budget with its 3DGS representation.

4.1 Experimental protocol

Datasets and splits. The benchmark comprises eight clinical volumes: six CT (abdomen, intestine, knee-joint, lower-limb, heart, vascular) and two MRI (nose, hand), spanning smooth and high-frequency anatomy. A physically based Monte Carlo volume path tracer produces a **general** set of 900 views (810 train/90 test). Camera distance varies; half of the views are intact and half contain a plane sampled across the volume core. Test-plane offsets are numerically disjoint from training and drawn from the same interval, evaluating interpolation to held-out cuts throughout the sampled volume core. A disjoint voxel-axis **cut-eval** set uses three fixed planes per volume and 30 cameras, each rendered both with and without clipping, yielding the paired references used by the localized metrics. A second set evaluates arbitrary-normal center planes without retraining. Supplement S3 specifies volume preparation, training-view sampling, and rendering; Supplement S5 specifies both cut-eval plane sets. The general set mixes fit-to-frame views with close-ups: intact views supply the outer anatomy and whole-volume context, while clipped cameras target the exposed face from oblique angles. The train/test split is stratified by view mode and plane axis, so each regime is represented without reusing an exact offset.

Comparison methods. **Ours** is the analytic half-space clip of Section 3.2. **MM** (forward-KL-optimal Gaussian surrogate) and **HC** (bare per-primitive hard cull) share Ours’ N-DGS backbone, initialization, planes, and supervision, isolating the operator within one representation. HC corresponds to the binary operator class without ClipGS’s STE training or deformation network. **ClipGS** (Li et al. 2025a) is our best-effort system-level reimplementation from its published description, using its vanilla-3DGS backbone, trainable visibility, and plane-conditioned deformation MLP; no public code was available. **RaRa** (Li et al. 2025b) enters only as a render-time operator on fixed interiors (Section 4.4). Supplement S4 details both baselines. Ours, MM,

HC, and ClipGS are trained for 30,000 iterations, and every result uses the fixed final checkpoint.

Evaluation metrics. On general views we report PSNR, SSIM (Wang et al. 2004), LPIPS (Zhang et al. 2018), and FPS. Cut-eval adds four cut-face metrics. S_b is band SSIM over the exposed cut face. The *difference-referenced cut error* CDE measures *where* the cut acts: for the method and reference, we subtract the clipped render from its unclipped companion, threshold the resulting removal map, and normalize the symmetric-difference area by the reference region. This construction cancels reconstruction error common to both renders and penalizes both missed cuts and over-removal. We report it over the exposed face (CDE_p) and grazing edge (CDE_g). Leak is the fraction of near-plane foreground energy on the culled side, where the reference is empty. As an image-space reference metric, it measures agreement with the independent path-traced reference and also captures proxy and renderer mismatch. Supplement S5 gives the equations, per-volume results, parameter sensitivity, and the camera-independent wrong-side-mass diagnostic $CErr_{3D}$, which measures operator geometry and is zero for Ours by analytic definition. Band PSNR differs by only hundredths and is omitted.

4.2 Held-out quality and speed

Image quality. Table 1 reports held-out quality and speed. Our method has the highest PSNR on every volume and improves over the ClipGS reimplementation by +1.22 dB on average PSNR. This is a system-level comparison because ClipGS uses the vanilla 3DGS representation specified in its paper. The controlled N-DGS ablations trace the operator effect: the smooth surrogate (MM, 33.29 dB) and the bare hard cull (HC, 33.14) both trail Ours (33.56), with the same overall trend in SSIM and LPIPS. Because every evaluation offset is disjoint from training, these scores show that the trained system maintains fidelity when interpolating to numerically held-out cross-sections along the sampled axes. The advantage also holds across all eight volumes rather than being driven by one scan: Ours leads PSNR in every row, from smooth soft tissue to thin vessels and fine joint structure. Global metrics nevertheless dilute the operator effect, because half of the held-out views contain no cut and, in the remaining views, the boundary occupies only a small image region. The largest Ours–HC gap occurs on *knee-joint*

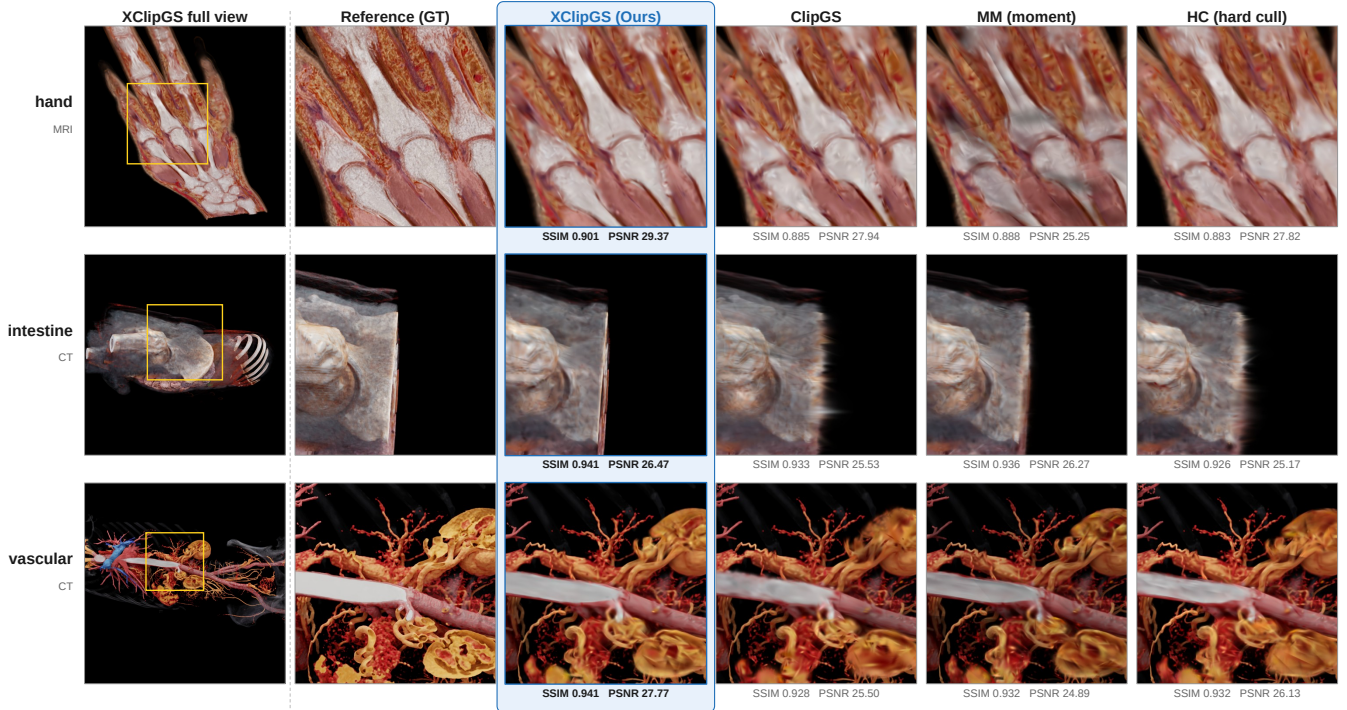


Figure 3: **Held-out qualitative comparison.** Full views locate each crop; the middle row is an unseen arbitrary-normal grazing view. The remaining columns compare matched reference and method crops, with SSIM and PSNR shown below.

(+0.70 dB), which contains fine cut-face structure. This dilution motivates the localized evaluation in Section 4.3.

Rendering speed. The analytic cut adds little measured overhead: Ours renders at 655 FPS, matching the hard cull (HC, 658) that does no per-sample work and the moment-matched surrogate (MM, 631), with all three within a few percent. Thus, the closed-form half-space factor adds negligible cost at this resolution on the measured A100. The ClipGS reimplementation runs at 278 FPS ($\sim 2.4\times$ slower), reflecting its extra plane-conditioned network per frame. The controlled Ours/MM/HC comparison shows the analytic operator itself runs at approximately hard-cull speed rather than trading fidelity for interactivity.

4.3 Cut-face fidelity and localization

Cut-eval protocol. To measure the clip where the global metric dilutes it, the cut-face metrics act on the held-out cut-eval set: three fixed planes per volume (one per voxel axis) at the volume core, each viewed by five *perpendicular* cameras (looking down the normal, so the cut face fills the frame) and five *grazing* cameras (tilted 1.5° off edge-on, exposing the cut edge). The clipped ground truth is rendered by the reference volume renderer with the plane applied by the renderer itself, and each camera also renders an *unclipped* companion for the difference-referenced error, supplying a matched volumetric reference for the partial-clipping behavior prior work calls “hard to define” (Li et al. 2025b) (details in Supplement S5). To test orientation generalization, a second set samples five arbitrary physical- world normals per volume, each at least 20° from every voxel axis and from the other

sampled normals. Each plane passes through the volume center and is viewed once perpendicularly and twice at grazing incidence, giving 15 views per volume. We evaluate the same checkpoints without retraining.

Image-space results. Table 2 and Figure 3 expose differences diluted by global metrics. XClipGS has the best average on all four metrics for both plane sets and the highest voxel-axis band SSIM on seven volumes. Without retraining, its arbitrary-normal averages are .874 S_b , 23.5 CDE_g , 14.3 CDE_p , and 0.13×10^{-2} Leak. It also has the highest displayed SSIM and PSNR across the examples, which span fine MRI structure in *hand*, layered soft tissue in *intestine*, and thin vessels in *vascular*. XClipGS preserves their structure and contrast while avoiding the diffuse or jagged boundaries of the approximations. Supplement S5 gives the per-volume results and arbitrary-normal $CErr_{3D}$.

Because these orientations and camera-plane configurations never appear in training, this set tests whether axis-supervised interiors support new center-plane orientations. The preserved ordering also shows that the analytic operator is not tied to voxel axes.

On voxel-axis cuts, Ours reduces CDE_g to 22.7, versus 23.6 for MM, 29.4 for HC, and 33.4 for ClipGS; Leak falls to 0.40×10^{-2} , versus 6.18, 12.54, and 16.42. MM retains an unbounded tail, while HC keeps or removes whole straddling splats. Supplement S5 confirms the lowest face CDE at all four thresholds and Leak in all 27 settings; Ours wins edge CDE in 11 of 16 settings and MM in five.

Table 2: **Average cut-face results over eight volumes.** S_b : band SSIM; CDE_g/CDE_p : edge/face error; Leak: leakage. Errors are $\times 10^{-2}$; **bold**: best per row.

Plane set	XClipGS (Ours)				ClipGS (reimpl.) (2025a)				MM				HC			
	$S_b \uparrow$	$CDE_g \downarrow$	$CDE_p \downarrow$	Leak $\downarrow$	$S_b \uparrow$	$CDE_g \downarrow$	$CDE_p \downarrow$	Leak $\downarrow$	$S_b \uparrow$	$CDE_g \downarrow$	$CDE_p \downarrow$	Leak $\downarrow$	$S_b \uparrow$	$CDE_g \downarrow$	$CDE_p \downarrow$	Leak $\downarrow$
Voxel-axis	.860	22.7	17.9	0.40	.809	33.4	19.5	16.42	.853	23.6	18.4	6.18	.819	29.4	18.5	12.54
Arbitrary-normal	.874	23.5	14.3	0.13	.844	35.9	15.7	17.24	.870	25.5	15.1	5.67	.842	35.9	14.9	18.30

Table 3: **Fixed-interior operator swap.** Each row changes only the clip rule on the voxel-axis set; metrics follow Table 2. **Bold**: full-precision row best; per-volume: Supplement S7.

Fixed interior	Ours (analytic)				RaRa (2025b)				MM				HC			
	$S_b \uparrow$	$CDE_g \downarrow$	$CDE_p \downarrow$	Leak $\downarrow$	$S_b \uparrow$	$CDE_g \downarrow$	$CDE_p \downarrow$	Leak $\downarrow$	$S_b \uparrow$	$CDE_g \downarrow$	$CDE_p \downarrow$	Leak $\downarrow$	$S_b \uparrow$	$CDE_g \downarrow$	$CDE_p \downarrow$	Leak $\downarrow$
Ours-trained	.860	22.7	17.9	0.40	.856	23.4	18.3	6.24	.848	23.7	18.0	5.85	.838	30.1	18.2	14.89
HC-trained	.828	23.1	18.5	0.37	.826	23.6	18.8	4.89	.828	23.5	18.4	5.12	.819	29.4	18.5	12.54

Global versus geometric fidelity. MM nearly matches Ours in global PSNR and SSIM, yet its unbounded tail produces about $15.5\times$ more leakage; HC is inexpensive but raises edge error. Our factor preserves the ordinary footprint and changes only the half-space mass, explaining its fidelity, speed, and cleaner boundary. Exactness is per-splat and operator-level: Equation 5 integrates each represented Gaussian inside the half-space, while projection, ordering, and compositing remain inherited from the base renderer. $CERR_{3D}$ is zero by construction even for arbitrary normals. Image-space Leak instead compares the learned proxy with an independent path-traced reference, so its small residual reflects density, color, and compositing mismatch rather than outside-plane mass from our operator.

4.4 Fixed-interior operator comparison

Operator-swap protocol. We render fixed checkpoints through each rule on the voxel-axis set, decoupling operator and interior. This admits RaRa (Li et al. 2025b), whose released rule lacks a backward path; its renderer agrees with ours to 42.8 dB on unclipped views, indicating small off-plane differences. The jointly trained ClipGS cull and deformation network remain in the system-level comparison above, where the complete learned system is evaluated.

Operator-swap results. On the Ours-trained interior, our operator has the best average CDE_g/CDE_p (22.7/17.9) and lowest Leak (0.40, versus 6.24 for RaRa, 5.85 for MM, and 14.89×10^{-2} for HC; Table 3). RaRa nearly matches band quality but leaks about $16\times$ more. This agrees with the analytic distinction: a chord fraction is not the Gaussian mass retained along the ray, whereas Equation 5 evaluates that mass in closed form at every covered pixel.

Table 2 additionally evaluates MM and HC on their co-adapted interiors, while the fixed-interior test supplies RaRa with the same clip-aware interior as the other rules. Repeating the swap on the HC-trained checkpoint preserves Ours’ average CDE_g and Leak lead across both interiors. Supplement S7 gives per-volume results; Supplement S4 details renderer agreement.

4.5 Clip-aware supervision ablation

We fix the clip rule and vary only supervision. The *external-view-only* control shares the backbone, initialization, test

split, and 30,000-iteration budget but trains only on intact views. It gains 0.7 dB on intact views yet loses 8.3 dB on every volume’s clipped views ($32.0 \rightarrow 23.8$). Voxel-axis band SSIM falls from .860 to .346 and CDE_p rises from 17.9 to 25.6×10^{-2} . The 8.3 dB gain from clip-aware supervision therefore quantifies its contribution to interior fidelity.

Together, the two controls isolate complementary effects: the operator swap attributes lower leakage to analytic clipping, while the supervision ablation holds that operator fixed and attributes interior fidelity to clipped views. Faithful clipping needs both (Supplement S6, Table S5).

4.6 Discussion

XClipGS separates responsibilities: clip-aware supervision learns the exposed anatomy, while the analytic renderer enforces each requested half-space. At deployment, a plane changes only transient per-primitive coefficients and one conditional-CDF factor per covered sample; one checkpoint therefore supports moving voxel-axis or oblique cuts without a plane-conditioned network or retraining. This separation also keeps the viewer interface simple: intact and clipped rendering share one learned volume and appearance model while the user updates only the requested plane.

5 Conclusion

XClipGS couples an exact closed-form per-splat affine-EWA integral with photometric interior supervision through the same differentiable operator, without learned clipping parameters or an auxiliary network. Across eight CT and MRI volumes, it leads all four cut-face metrics on voxel-axis and arbitrary-normal planes at hard-cull speed. The 8.3 dB advantage from clipped training views confirms that faithful clipping benefits from both analytic truncation and a supervised interior. Paired clipped/unclipped evaluation localizes boundary error, while arbitrary-normal tests demonstrate transfer beyond the supervised axes. Because new half-spaces require no retraining, XClipGS provides a reusable primitive for interactive Gaussian volume viewers.

References

- Arellano-Valle, R. B.; Branco, M. D.; and Genton, M. G. 2006. A Unified View on Skewed Distributions Arising from Selections. *Canadian Journal of Statistics*, 34(4): 581–601.
- Barron, J. T.; Mildenhall, B.; Tancik, M.; Hedman, P.; Martin-Brualla, R.; and Srinivasan, P. P. 2021. Mip-NeRF: A Multiscale Representation for Anti-Aliasing Neural Radiance Fields. In *ICCV*, 5855–5864.
- Cai, Y.; Liang, Y.; Wang, J.; Wang, A.; Zhang, Y.; Yang, X.; Zhou, Z.; and Yuille, A. 2024. Radiative Gaussian Splatting for Efficient X-ray Novel View Synthesis. In *ECCV*.
- Cen, J.; Fang, J.; Yang, C.; Xie, L.; Zhang, X.; Shen, W.; and Tian, Q. 2025. Segment Any 3D Gaussians. In *AAAI*.
- Chen, Y.; Chen, Z.; Zhang, C.; Wang, F.; Yang, X.; Wang, Y.; Cai, Z.; Yang, L.; Liu, H.; and Lin, G. 2024. GaussianEditor: Swift and Controllable 3D Editing with Gaussian Splatting. In *CVPR*.
- Comaniciu, D.; Engel, K.; Georgescu, B.; and Mansi, T. 2016. Shaping the future through innovations: From medical imaging to precision medicine. *Medical Image Analysis*, 33: 19–26.
- Dappa, E.; Higashigaito, K.; Fornaro, J.; Leschka, S.; Wildermuth, S.; and Alkadhi, H. 2016. Cinematic rendering – an alternative to volume rendering for 3D computed tomography imaging. *Insights into Imaging*, 7(6): 849–856.
- Feng, Q.-Y.; Cao, G.-C.; Chen, H.-X.; Xu, Q.-C.; Mu, T.-J.; Martin, R. R.; and Hu, S.-M. 2024. EVSplitting: An Efficient and Visually Consistent Splitting Algorithm for 3D Gaussian Splatting. In *SIGGRAPH Asia 2024 Conference Papers*, 1–11.
- Gao, Z.; Planche, B.; Zheng, M.; Chen, X.; Chen, T.; and Wu, Z. 2024. DDGS-CT: Direction-Disentangled Gaussian Splatting for Realistic Volume Rendering. *Advances in Neural Information Processing Systems*.
- Gao, Z.; Planche, B.; Zheng, M.; Choudhuri, A.; Chen, T.; and Wu, Z. 2025a. 6dgs: Enhanced direction-aware gaussian splatting for volumetric rendering. In *ICLR*.
- Gao, Z.; Planche, B.; Zheng, M.; Choudhuri, A.; Chen, T.; and Wu, Z. 2025b. 7DGS: Unified Spatial-Temporal-Angular Gaussian Splatting. In *ICCV*.
- Gao, Z.; Planche, B.; Zheng, M.; Choudhuri, A.; Nguyen, V. N.; Chen, T.; and Wu, Z. 2026. Render-FM: Feedforward Model for Real-time Photorealistic Volumetric Rendering. In *ECCV*. ArXiv:2505.17338.
- Hahlbohm, F.; Franke, L.; Eisemann, M.; and Magnor, M. 2026. Faster-GS: Analyzing and Improving Gaussian Splatting Optimization. In *CVPR*.
- Hanson, A.; Tu, A.; Lin, G.; Singla, V.; Zwicker, M.; and Goldstein, T. 2025. Speedy-Splat: Fast 3D Gaussian Splatting with Sparse Pixels and Sparse Primitives. In *CVPR*, 21537–21546.
- Huang, Z.; Wu, C.-Y.; Guo, Y.; Huang, X.; and Ren, L. 2026. 3DGEER: 3D Gaussian Rendering Made Exact and Efficient for Generic Cameras. In *ICLR*.
- Jain, U.; Mirzaei, A.; and Gilitschenski, I. 2024. Gaussian-Cut: Interactive Segmentation via Graph Cut for 3D Gaussian Splatting. In *NeurIPS*.
- Kerbl, B.; Kopanas, G.; Leimkühler, T.; and Drettakis, G. 2023. 3D Gaussian Splatting for Real-Time Radiance Field Rendering. *ACM Transactions on Graphics*, 42(4): 139:1–139:14.
- Kleinbeck, C.; Schieber, H.; Engel, K.; Gutjahr, R.; and Roth, D. 2025. Multi-Layer Gaussian Splatting for Immersive Anatomy Visualization. *IEEE Transactions on Visualization and Computer Graphics*, 31(5): 2353–2363.
- Kroes, T.; Post, F. H.; and Botha, C. P. 2012. Exposure Render: An Interactive Photo-Realistic Volume Rendering Framework. *PLoS ONE*, 7(7): e38586.
- Li, C.; Tong, Y.; Chen, K.; Yang, Z.; Li, R.; Qiu, S.; Chan, J. Y.-K.; Heng, P.-A.; and Dou, Q. 2025a. ClipGS: Clippable Gaussian Splatting for Interactive Cinematic Visualization of Volumetric Medical Data. In *MICCAI*.
- Li, D.; Jia, D.; Rajeh, Y.; Engel, D.; and Viola, I. 2025b. RaRa Clipper: A Clipper for Gaussian Splatting Based on Ray Tracer and Rasterizer. In *SIGGRAPH Asia 2025 Conference Papers*.
- Li, H.; Liu, J.; Sznai, M.; and Camps, O. 2025c. 3D-HGS: 3D Half-Gaussian Splatting. In *CVPR*, 10996–11005.
- Liang, Z.; Zhang, Q.; Hu, W.; Zhu, L.; Feng, Y.; and Jia, K. 2024. Analytic-Splatting: Anti-Aliased 3D Gaussian Splatting via Analytic Integration. In *ECCV*.
- Liao, Z.; Ding, J.; Cui, S.; Gong, R.; Hu, B.; Wang, Y.; Li, H.; Wang, H.; Zhang, X.; and Fu, R. 2025. TC-GS: A Faster Gaussian Splatting Module Utilizing Tensor Cores. In *SIGGRAPH Asia 2025 Conference Papers*, 181:1–181:9.
- Liu, R.; Gao, Z.; Planche, B.; Chen, M.; Nguyen, V. N.; Zheng, M.; Choudhuri, A.; Chen, T.; Wang, Y.; Feng, A.; and Wu, Z. 2026. Universal Beta Splatting. In *ICLR*. ArXiv:2510.03312.
- Liu, R.; Sun, D.; Chen, M.; Wang, Y.; and Feng, A. 2025. Deformable beta splatting. In *SIGGRAPH*.
- Martschinke, J.; Hartnagel, S.; Keinert, B.; Engel, K.; and Stamminger, M. 2019. Adaptive Temporal Sampling for Volumetric Path Tracing of Medical Data. *Computer Graphics Forum*, 38(4): 67–76.
- Niedermayr, S.; Neuhauser, C.; Petkov, K.; Engel, K.; and Westermann, R. 2024. Application of 3D Gaussian Splatting for Cinematic Anatomy on Consumer Class Devices. In *Vision, Modeling, and Visualization (VMV)*. The Eurographics Association. ArXiv:2404.11285.
- Novák, J.; Georgiev, I.; Hanika, J.; and Jarosz, W. 2018. Monte Carlo Methods for Volumetric Light Transport Simulation. *Computer Graphics Forum*, 37(2): 551–576. Proc. Eurographics State of the Art Reports.
- Peng, T.; Zha, R.; Li, Z.; Liu, X.; and Zou, Q. 2025. Three-Dimensional MRI Reconstruction with Gaussian Representations: Tackling the Undersampling Problem. *arXiv preprint arXiv:2502.06510*.
- Tallis, G. M. 1961. The moment generating function of the truncated multi-normal distribution. *Journal of the Royal*

Statistical Society: Series B (Methodological), 23(1): 223–229.

Wang, Z.; Bovik, A. C.; Sheikh, H. R.; and Simoncelli, E. P. 2004. Image Quality Assessment: From Error Visibility to Structural Similarity. *IEEE Transactions on Image Processing*, 13(4): 600–612.

Weiskopf, D.; Engel, K.; and Ertl, T. 2003. Interactive Clipping Techniques for Texture-Based Volume Visualization and Volume Shading. *IEEE Transactions on Visualization and Computer Graphics*, 9(3): 298–312.

Xu, J.; Thevenon, G.; Chabat, T.; McCormick, M.; Li, F.; Birdsong, T.; Martin, K.; Lee, Y.; and Aylward, S. 2023. Interactive, in-browser cinematic volume rendering of medical images. *Computational Methods in Biomechanics and Biomedical Engineering: Imaging & Visualization*, 11(4): 1019–1026.

Yang, G.; Kieß, S.; Luo, H.; Liu, X.; and Simon, S. 2026. Exact-GS: Mathematically Rigorous and Accurate 3D Gaussian Splatting for 3D X-ray Reconstruction. In *CVPR*, 4902–4911.

Yang, S.; Li, Q.; Shen, D.; Gong, B.; Dou, Q.; and Jin, Y. 2024. Deform3DGS: Flexible Deformation for Fast Surgical Scene Reconstruction with Gaussian Splatting. In *MICCAI*.

Ye, M.; Danelljan, M.; Yu, F.; and Ke, L. 2024. Gaussian Grouping: Segment and Edit Anything in 3D Scenes. In *ECCV*.

Yu, Z.; Chen, A.; Huang, B.; Sattler, T.; and Geiger, A. 2024. Mip-Splatting: Alias-free 3D Gaussian Splatting. In *CVPR*.

Zha, R.; Lin, T. J.; Cai, Y.; Cao, J.; Zhang, Y.; and Li, H. 2024. R2-Gaussian: Rectifying Radiative Gaussian Splatting for Tomographic Reconstruction. In *NeurIPS*.

Zhang, R.; Isola, P.; Efros, A. A.; Shechtman, E.; and Wang, O. 2018. The Unreasonable Effectiveness of Deep Features as a Perceptual Metric. In *CVPR*, 586–595.

Zhou, S.; Chang, H.; Jiang, S.; Fan, Z.; Zhu, Z.; Xu, D.; Chari, P.; You, S.; Wang, Z.; and Kadambi, A. 2024. Feature 3DGS: Supercharging 3D Gaussian Splatting to Enable Distilled Feature Fields. In *CVPR*.

Zwicker, M.; Pfister, H.; Van Baar, J.; and Gross, M. 2001. EWA volume splatting. In *Proceedings Visualization, 2001. VIS'01.*, 29–36. IEEE.

Supplementary Material

XClipGS: Exact Half-Space Clipping for Medical Volume Gaussian Splatting

This supplement collects the derivations, protocol, and per-scan results that the main paper refers to by section number. Section S1 gives the full proof of the exactness proposition; Section S2 details the CUDA rasterizer forward/backward and its numerical validation; Section S3 specifies the benchmark volumes, view sampling, and training and timing setup; Section S4 describes the ClipGS and RaRa baseline implementations; Section S5 defines the cut-face metrics (S_b , CDE, Leak, CErr_{3D}) and tests their parameter sensitivity; Section S6 reports the external-view-only ablation per scan; Section S7 gives the operator-swap controls and per-volume tables; and Section S8 describes the video and interactive demonstrations available on the project page. All numbers are produced by the same pipeline as the main paper and reuse its notation.

S1 Proof of Analytic Exactness

We give the full proof of Proposition 1 in the main paper. Fix a primitive $(\boldsymbol{\mu}, \boldsymbol{\Sigma}, \alpha)$ with $\boldsymbol{\Sigma} \succ 0$ and a camera with the local affine model $\mathbf{u}(\mathbf{x}) = \hat{\mathbf{p}} + \mathbf{M}(\mathbf{x} - \boldsymbol{\mu})$, $\mathbf{M} \in \mathbb{R}^{2 \times 3}$ of rank 2, and let $\omega(\mathbf{x}) = \mathbf{n}^\top \mathbf{x}$ be the clip coordinate.

An affine chart adapted to the rays. Under the affine model, the viewing ray of pixel $\mathbf{p}$ is the line $\{\mathbf{x} : \mathbf{u}(\mathbf{x}) = \mathbf{p}\}$, whose direction spans the one-dimensional kernel of $\mathbf{M}$. Choose a unit vector $\mathbf{d}$ with $\mathbf{M}\mathbf{d} = \mathbf{0}$ and set $t(\mathbf{x}) = \mathbf{d}^\top (\mathbf{x} - \boldsymbol{\mu})$. The map $T : \mathbf{x} \mapsto (\mathbf{u}(\mathbf{x}), t(\mathbf{x}))$ is affine, and it is invertible: if $\mathbf{M}\mathbf{v} = \mathbf{0}$ then $\mathbf{v} = c\mathbf{d}$, and $\mathbf{d}^\top \mathbf{v} = c = 0$. Its Jacobian $J = |\det[\mathbf{M}; \mathbf{d}^\top]|$ is a nonzero constant. Hence if $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with density ρ , the pair $(\mathbf{u}, t) = T(\mathbf{x})$ is jointly Gaussian with density $f(\mathbf{u}, t) = J^{-1} \rho(T^{-1}(\mathbf{u}, t))$, and ω , being another affine image of $\mathbf{x}$, is jointly Gaussian with $(\mathbf{u}, t)$.

Factorizing the clipped ray integral. The reference contribution of the primitive to pixel $\mathbf{p}$ under the half-space restriction is the ray integral of the restricted density,

$$I_{\text{clip}}(\mathbf{p}) = \alpha \int_{\mathbb{R}} \rho(\mathbf{x}(\mathbf{p}, t)) \mathbb{1}[\omega(\mathbf{x}(\mathbf{p}, t)) \leq \tau] dt, \quad (\text{S1})$$

where $\mathbf{x}(\mathbf{p}, t) = T^{-1}(\mathbf{p}, t)$ traverses the ray. Writing the integrand through f and factorizing the joint density into marginal and conditional, $f(\mathbf{p}, t) = f_U(\mathbf{p}) f_{T|U}(t | \mathbf{p})$,

$$I_{\text{clip}}(\mathbf{p}) = \alpha J f_U(\mathbf{p}) \int_{\mathbb{R}} f_{T|U}(t | \mathbf{p}) \mathbb{1}[\omega \leq \tau] dt \quad (\text{S2})$$

$$= \alpha J f_U(\mathbf{p}) P(\omega \leq \tau | \mathbf{u} = \mathbf{p}), \quad (\text{S3})$$

because, given $\mathbf{u} = \mathbf{p}$, the clip coordinate ω is an affine function of t alone, so integrating the conditional density of t over the event $\{\omega \leq \tau\}$ is exactly its conditional probability. Setting $\tau = \infty$ recovers the unclipped integral $I_{\text{full}}(\mathbf{p}) = \alpha J f_U(\mathbf{p})$, so

$$I_{\text{clip}}(\mathbf{p}) = I_{\text{full}}(\mathbf{p}) \cdot P(\omega \leq \tau | \mathbf{u} = \mathbf{p}). \quad (\text{S4})$$

The marginal f_U is the Gaussian $\mathcal{N}(\mathbf{p}; \hat{\mathbf{p}}, \mathbf{A})$ with $\mathbf{A} = \mathbf{M}\boldsymbol{\Sigma}\mathbf{M}^\top$, which is positive definite because $\boldsymbol{\Sigma} \succ 0$ and $\mathbf{M}$ has full row rank. Up to the constant $2\pi|\mathbf{A}|^{1/2}$ this is the unnormalized EWA footprint $\mathcal{K}_G(\boldsymbol{\delta}; \mathbf{A})$ with $\boldsymbol{\delta} = \hat{\mathbf{p}} - \mathbf{p}$; the splatting kernel absorbs this constant and J identically in the clipped and unclipped cases, which is the “global normalization shared with the unclipped splat” in the proposition. Note also that the result is independent of the choice of $\mathbf{d}$: any other affine ray coordinate rescales J and $f_{T|U}$ jointly and leaves Equation (S4) unchanged.

The conditional law of the clip coordinate. The pair $(\omega, \mathbf{u})$ is jointly Gaussian with

$$E[\omega] = \mathbf{n}^\top \boldsymbol{\mu}, \quad \text{Var}(\omega) = \mathbf{n}^\top \boldsymbol{\Sigma} \mathbf{n} = \sigma_n^2, \quad (\text{S5})$$

$$\text{Cov}(\omega, \mathbf{u}) = \mathbf{n}^\top \boldsymbol{\Sigma} \mathbf{M}^\top = \mathbf{b}^\top, \quad (\text{S6})$$

where $\mathbf{b} = \mathbf{M}\boldsymbol{\Sigma}\mathbf{n}$. The standard Gaussian conditioning formulas give

$$\begin{aligned} E[\omega | \mathbf{u} = \mathbf{p}] &= \mathbf{n}^\top \boldsymbol{\mu} + \mathbf{b}^\top \mathbf{A}^{-1}(\mathbf{p} - \hat{\mathbf{p}}) \\ &= \mathbf{n}^\top \boldsymbol{\mu} - \mathbf{b}^\top \mathbf{A}^{-1} \boldsymbol{\delta}, \end{aligned} \quad (\text{S7})$$

$$\text{Var}(\omega | \mathbf{u} = \mathbf{p}) = \sigma_n^2 - \mathbf{b}^\top \mathbf{A}^{-1} \mathbf{b} = s^2, \quad (\text{S8})$$

the Schur complement of $\mathbf{A}$ in the joint covariance of $(\omega, \mathbf{u})$. For $s > 0$, standardizing yields

$$\begin{aligned} P(\omega \leq \tau | \mathbf{u} = \mathbf{p}) &= \Phi\left(\frac{\tau - \mathbf{n}^\top \boldsymbol{\mu} + \mathbf{b}^\top \mathbf{A}^{-1} \boldsymbol{\delta}}{s}\right) \\ &= \Phi(\mathbf{k}^\top \boldsymbol{\delta} + h), \end{aligned} \quad (\text{S9})$$

with $\mathbf{k} = \mathbf{A}^{-1} \mathbf{b} / s$ and $h = (\tau - \mathbf{n}^\top \boldsymbol{\mu}) / s$, using the symmetry of $\mathbf{A}^{-1}$. Substituting into Equation (S4) gives the clipped per-pixel contribution $\alpha \mathcal{K}_G(\boldsymbol{\delta}; \mathbf{A}) \Phi(\mathbf{k}^\top \boldsymbol{\delta} + h)$, which is Equation 5 of the main paper. ■

Degenerate cases. $s^2 \geq 0$ always, with $s^2 = 0$ exactly when ω is almost surely an affine function of $\mathbf{u}$, i.e., when the clip coordinate is determined by the pixel; geometrically the plane then contains the viewing-ray direction. In that limit the conditional law collapses to a point mass and $\Phi(\ell) \rightarrow \mathbb{1}[E[\omega | \mathbf{u}] \leq \tau]$, the per-pixel step described in the main paper. The finite-precision kernel handles this regime with the conditional-variance floor documented in Section S2.

The operator ladder. In the same sense, the three operators of the main paper render ray integrals of the same restricted density r under successively weaker projections of it: MM renders the ray integral of the Gaussian moment projection of r , and HC the ray integral of r ’s binary keep/drop projection. Comparing them therefore isolates operator fidelity on a single ladder.

Scope of exactness. The proof is per primitive: it establishes the exact retained ray integral relative to that primitive’s affine-EWA footprint. The complete rasterizer continues to use the base renderer’s center-based splat order and front-to-back alpha compositing. The proposition thus isolates the added half-space integral, while the complete image retains the established ordering and compositing approximations of the base EWA rasterizer.

S2 Rasterizer Implementation and Validation

Forward pass. HC, MM, and Ours share a tile-based CUDA rasterizer that combines Speedy-Splat’s tight tile culling (Hanson et al. 2025), the packed half-precision tensor-core forward of TC-GS (Liao et al. 2025), and the Mip-Splatting 3D low-pass filter (Yu et al. 2024). Before rasterization, MM and Ours discard primitives with negligible kept mass $\Phi(a) \leq 10^{-6}$. MM also applies the $1/255$ compositing threshold to its mass-scaled opacity $\alpha\Phi(a)$. MM rewrites (μ, Σ, α) during preprocess, whereas Ours computes the three clip coefficients and multiplies sample opacity by $\Phi(\ell)$ only for plane-straddling primitives. To avoid division by zero near the degenerate limit, the implementation uses $s_{\text{impl}}^2 = \max(s^2, 10^{-8})$ and evaluates the CDF in fp32. The backward pass sets the derivative through s^2 to zero while this floor is active; this branch is evaluated only for clipped scenes. Because ℓ is affine in the pixel coordinate, the factor integrates into the packed forward with its coefficients staged per tile. The existing forward path is retained, with one scalar fp32 CDF evaluation added for each straddling sample. Here “straddling” names the finite-precision rasterizer’s active clipping branch. The 10^{-6} retained-mass cutoff and 10^{-8} variance floor provide its numerical safeguards, while Proposition 1 describes the corresponding ideal-Gaussian operator.

Backward pass. The backward pass emits gradients for the clip coefficients and projected center and chains them through the conic and projection Jacobian to (μ, Σ) . When the variance floor is active, its derivative with respect to the unclamped s^2 is set to zero; all other paths retain their finite analytic gradients. Training uses a warp-per-splat render backward in the style of Faster-GS (Hahlbohm et al. 2026), in which each lane accumulates one primitive’s full gradient, including its three clip coefficients, in registers and issues a single atomic per primitive per tile; on an A100 at 800×800 with 300k Gaussians and an active oblique clip, this reduces a measured forward-backward step from 5.47 to 3.50 ms ($1.6\times$) relative to the legacy per-pixel backward.

Numerical validation and overhead. We validate forward and backward passes against a float64 autograd implementation of the full pipeline. All parameter gradients match to approximately 10^{-6} relative error for axis-aligned and oblique planes, with clipping active and inactive. A separate robustness suite covers rotated camera poses, partial tiles at image borders, alpha saturation, early ray termination, and rotation equivariance of the clip (rotating the scene and the plane together leaves the render unchanged). Relative to MM, Ours requires one transient 16-byte per-view buffer per primitive and one CDF evaluation per straddling sample; it introduces no learned primitive parameters. These float64 tests provide independent numerical validation of the implemented formulas and safeguards; Section S5’s zero-by-construction diagnostic instead measures operator geometry.

S3 Benchmark and Training Setup

Volumes and reference renderer. The eight volumes are de-identified clinical CT and MRI scans; `nose` and `hand` are MRI, the other six are CT. Each is read from its DICOM series and rendered on its own voxel grid (for example, `heart` is $512 \times 512 \times 317$), with volumes kept on their native grids rather than resampled to a common resolution. For each scan, a fixed render preset specifies the intensity-to-RGBA transfer function, material parameters, environment and area lights, and Monte Carlo quality settings for every view. All presets use the `uffizi-large.hdr` environment map; their scan-specific progressive rendering limits range from 1,000 to 7,135 iterations. The cinematic, physically based path tracer is the reference throughout: all “ground truth” and clipped reference images are its output. Clipping uses its native volumetric half-space interface rather than a splatting operator. Thus clipped and unclipped renders from one camera share the volume, transfer function, lighting, materials, and camera and differ only by the half-space clip. The exact per-volume metadata, render preset, and renderer configuration will be released with the code. The benchmark measures rendering fidelity on these eight scans, each treated as one scene.

Views and sampling. Every image is rendered at 1600×1600 with a 60° vertical field of view. We generate each 900-view set once with random seed 7: 300 intact fit-to-frame views at distance factor 1.0, 300 clipped near-fit views at $[0.85, 1.0]$, 150 clipped close-ups at $[0.4, 0.6]$, and 150 intact close-ups at $[0.4, 0.6]$. For a clipped view, we sample one of the three voxel axes uniformly and an offset τ uniformly over the central 30%–70% of its spatial extent. The camera targets a point on the exposed face and is sampled 25° – 70° from the plane normal. The 810/90 train/test split is stratified within every view-mode and cut-axis combination; conversion preserves this split rather than sampling it again. Test offsets are therefore numerically disjoint from training offsets but come from the same offset interval, measuring interpolation to held-out cuts throughout the sampled volume core. The voxel-axis cut-evaluation set contains 30 cameras per volume, each producing clipped and unclipped renders on three fixed planes (one per voxel axis) at the volume-core midpoint. A separate orientation set uses five deterministic physical-world center planes per volume, with one perpendicular and two grazing views per plane. Both evaluation sets are held out from optimization, and the arbitrary-normal set uses the original checkpoints without retraining.

Coordinate alignment. After reference rendering, camera poses, clipping planes, and initialization points are transformed into one volume-centered physical coordinate frame. Each cut is stored as the retained half-space $\mathbf{n}^\top \mathbf{x} \leq \tau$. For an obliquely oriented DICOM grid, the selected voxel-axis normal and its offset are transformed into patient/world coordinates; the volume itself remains on its native grid. For the arbitrary-normal set, we instead sample a physical-world normal and center plane and convert it to the renderer’s index-coordinate plane equation. The initialization point cloud is centered once by the same volume-center translation, ensuring that the images, cameras, planes, and primitives share

one frame. In both evaluation sets, the reference renderer and splatting methods receive the same physical plane.

Timing. All FPS figures are measured on a single, otherwise-idle NVIDIA A100-SXM4-80GB at the 1600×1600 evaluation resolution. For each checkpoint we render all 90 held-out test views for three passes (270 frames), discard the first 30 as warm-up, time each remaining frame with `torch.cuda.synchronize` on both sides, and report the reciprocal of the *median* per-frame time; the median and the single-GPU run guard against transient contention. Timing includes the full per-view forward, i.e. opacity conditioning and the clip factor for our operator and the deformation-network evaluation for the ClipGS reimplementation; it excludes disk I/O and image saving. Consequently, the FPS comparison characterizes A100 performance at the stated resolution. Ours/MM/HC isolate operator cost in the shared renderer, whereas ClipGS timing remains a system-level comparison that includes its additional network.

Backbones and warm start. All four systems receive the same per-volume initialization, using Render-FM feed-forward predictions where available (Gao et al. 2026). The warm start is stored as a point cloud carrying positions, spherical-harmonic color, opacity, scale, rotation, and conditioning attributes; each backbone loads the attributes it supports. Ours, MM, and HC use N-DGS as the backbone (Gao et al. 2025a,b; Liu et al. 2026). The Cholesky-precision diagonal of the conditioning block is initialized at 2.0. The ClipGS reimplementation uses vanilla 3DGS, as in the cited paper (Li et al. 2025a).

Optimization. All systems train for 30,000 iterations with densification enabled, $\lambda_s = 0.2$ in the photometric loss, and otherwise the standard 3DGS optimizer settings. The Mip-Splatting filter is enabled for the three N-DGS variants; the ClipGS system baseline uses vanilla 3DGS without the filter. All tables and figures use the fixed 30,000-iteration checkpoint; the test set is reserved for final evaluation. Each training frame carries its clipping plane $(\mathbf{n}, \tau)$ in the dataset metadata, and the same plane is passed to the rasterizer during training. Each scan-method pair is trained once with the Python, NumPy, and PyTorch random generators fixed to seed 0; reported image metrics aggregate all held-out views from that fixed run. The tables therefore report deterministic, matched fixed-run comparisons; each metric aggregates all held-out views from that run.

S4 Baseline Implementation Details

ClipGS reimplementation. We implement ClipGS (Li et al. 2025a) from its published description, following the stated vanilla-3DGS backbone, trainable binary visibility, and plane-conditioned deformation MLP. We train it end to end under the same initialization, data, clipping planes, and iteration budget as the other systems. We report it as a system-level comparison; the controlled operator conclusions come from Ours, MM, and HC on the shared N-DGS backbone.

RaRa integration. We evaluate RaRa (Li et al. 2025b) through the authors’ released CUDA rasterization kernel,

driven by our cut-eval cameras and the shared fixed checkpoints. Their plane convention keeps the side $\mathbf{n}^\top \mathbf{x} + d > 0$, so our plane $(\mathbf{n}, \tau)$ is passed as $(-\mathbf{n}, \tau)$, and RaRa is rendered with the same Mip-Splatting-filtered forward as the other operators, matching antialiasing across the comparison. We applied three compatibility fixes to the released kernel before evaluation: the per-primitive decay-weight buffer is initialized to one, so that primitives classified as fully visible render unchanged rather than vanishing, and chords whose endpoints are both invisible contribute zero; we additionally removed a duplicated function specifier for toolchain compatibility. These fixes preserve the clipping formula. Rendering the identical unclipped checkpoint through both rasterizers at the cut-eval cameras gives 42.8 dB mean PSNR, confirming matched off-plane behavior.

S5 Cut-Eval Protocol and Metric Definitions

For the voxel-axis evaluation, we fix three clipping planes per volume, one per voxel axis at the volume core, and render two view families of five cameras per plane, giving 30 views per volume, all held out from training. Perpendicular views look down the plane normal so the exposed cut face fills the frame; grazing views look nearly along the plane, tilted by 1.5° to expose its cut edge. The clipped reference is produced by the reference volume renderer with its native half-space clip; the unclipped companion uses the same camera with clipping disabled.

Table S1 expands the voxel-axis average in the main paper into per-volume values. XClipGS has the lowest Leak on every volume and the highest S_b on seven of eight. Its Leak spans 0.02 on *lower-limb* to 1.45 on *knee-joint* (all $\times 10^{-2}$). This cross-volume range reflects the scene-dependent near-plane foreground energy in the metric denominator; within *knee-joint*, XClipGS remains below MM (14.65), HC (22.44), and ClipGS (28.64).

Arbitrary-normal generalization. The main benchmark varies plane offsets along the three voxel axes. To test orientation generalization separately, we sample five deterministic physical-world normals per volume, rejecting any normal within 20° of a voxel axis or within 20° of another sampled orientation. Sampling is performed once with seed 27. Every plane passes through the volume center and has one perpendicular and two grazing views, for 15 views per volume. We render matched clipped and unclipped references and evaluate the original checkpoints without retraining. Table S2 gives the complete results. XClipGS has the lowest Leak on all eight volumes, the lowest CDE_g on seven, and the highest S_b on six. MM has the lowest CDE_g on *vascular* and the lowest CDE_p on four volumes, but XClipGS remains best on the average of every reported metric.

For the near-edge coordinate in a grazing view, we use the deterministic least-squares line proxy $\mathbf{l} = \text{pinv}(\mathbf{P})^\top [\mathbf{n}; -\tau]$, where $\mathbf{P}$ is the camera projection matrix. Equivalently, $\mathbf{l}$ minimizes $\|\mathbf{P}^\top \mathbf{l} - [\mathbf{n}; -\tau]\|_2$. A general 3D plane projects nonlinearly under perspective, so this proxy supplies a stable near-edge coordinate for our cameras placed 1.5° from edge-on. Its sign is oriented so reference foreground adjacent to the cut lies at $s < 0$, independently of the stored plane

Table S1: **Per-volume cut-face evaluation.** Errors and Leak are $\times 10^{-2}$; **bold** marks the full-precision best.

Scene	XClipGS (Ours)				ClipGS				MM				HC			
	$S_b \uparrow$	$CDE_g \downarrow$	$CDE_p \downarrow$	Leak $\downarrow$	$S_b \uparrow$	$CDE_g \downarrow$	$CDE_p \downarrow$	Leak $\downarrow$	$S_b \uparrow$	$CDE_g \downarrow$	$CDE_p \downarrow$	Leak $\downarrow$	$S_b \uparrow$	$CDE_g \downarrow$	$CDE_p \downarrow$	Leak $\downarrow$
abdomen	.899	28.2	34.4	0.26	.890	43.1	39.8	12.93	.895	31.3	38.6	3.32	.895	38.9	34.7	10.50
intestine	.920	22.4	17.2	0.16	.901	30.9	17.7	12.68	.924	20.4	17.5	2.79	.902	25.6	17.7	8.28
knee-joint	.825	15.2	11.1	1.45	.707	24.7	12.2	28.64	.793	15.7	10.7	14.65	.734	19.6	11.6	22.44
lower-limb	.875	19.2	19.0	0.02	.821	30.5	19.9	9.81	.869	19.4	18.8	2.09	.847	26.0	19.6	8.08
vascular	.876	25.4	13.1	0.06	.838	29.1	14.1	5.19	.868	24.9	14.2	0.77	.847	27.9	13.6	3.79
heart	.877	21.5	12.2	0.41	.824	32.5	12.9	30.68	.871	24.5	11.8	16.33	.826	29.2	12.4	23.00
nose(MRI)	.720	25.3	15.2	0.09	.646	39.6	16.0	17.29	.718	28.3	15.0	3.88	.665	34.0	15.6	9.62
hand(MRI)	.886	24.6	20.9	0.71	.848	36.3	23.2	14.12	.883	24.2	20.7	5.60	.838	34.1	22.7	14.63
average	.860	22.7	17.9	0.40	.809	33.4	19.5	16.42	.853	23.6	18.4	6.18	.819	29.4	18.5	12.54

Table S2: **Per-volume arbitrary-normal evaluation.** Errors and Leak are $\times 10^{-2}$; **bold** marks the full-precision best.

Scene	XClipGS (Ours)				ClipGS				MM				HC			
	$S_b \uparrow$	$CDE_g \downarrow$	$CDE_p \downarrow$	Leak $\downarrow$	$S_b \uparrow$	$CDE_g \downarrow$	$CDE_p \downarrow$	Leak $\downarrow$	$S_b \uparrow$	$CDE_g \downarrow$	$CDE_p \downarrow$	Leak $\downarrow$	$S_b \uparrow$	$CDE_g \downarrow$	$CDE_p \downarrow$	Leak $\downarrow$
abdomen	.924	24.9	26.3	0.27	.912	39.3	31.7	15.62	.922	27.8	28.7	3.80	.901	39.0	27.4	17.52
intestine	.918	18.6	12.7	0.11	.914	31.5	13.1	14.43	.925	22.3	12.9	2.94	.904	31.6	13.1	16.14
knee-joint	.901	18.8	10.0	0.16	.849	32.0	11.8	28.64	.884	22.2	9.8	12.90	.852	32.2	10.7	31.06
lower-limb	.903	19.5	8.7	0.02	.884	29.5	9.7	8.95	.907	20.3	8.6	2.02	.899	29.3	8.7	9.67
vascular	.881	34.7	9.2	0.03	.847	35.9	10.5	5.27	.878	26.0	14.0	0.98	.860	35.3	9.6	5.75
heart	.846	21.8	14.2	0.05	.797	28.1	14.7	26.00	.841	22.1	13.8	11.65	.805	30.3	14.5	29.44
nose(MRI)	.806	24.9	17.2	0.11	.755	44.5	17.8	17.88	.790	30.3	17.3	4.45	.747	44.2	18.1	17.61
hand(MRI)	.815	24.2	16.4	0.26	.790	46.5	16.6	21.12	.812	32.7	15.9	6.58	.769	45.7	16.8	19.18
average	.874	23.5	14.3	0.13	.844	35.9	15.7	17.24	.870	25.5	15.1	5.67	.842	35.9	14.9	18.30

orientation. Grazing metrics use a ± 12 -pixel strip around this line; perpendicular metrics use the exposed-face foreground.

Notation. All image metrics operate on scalar luminance $Y(\mathbf{x}) = \frac{1}{3} \sum_c I_c(\mathbf{x}) \in [0, 1]$ (mean of the three $[0, 1]$ channels) at pixel $\mathbf{x}$. The reference renderer uses a black background, so the foreground mask is $\mathcal{F}(I) = \{\mathbf{x} : \max_c I_c(\mathbf{x}) > \theta_{fg}\}$ with $\theta_{fg} = 0.04$. Write $s(\mathbf{x})$ for the signed pixel distance to the near-edge line proxy, oriented so $s < 0$ is the kept side. Two band restrictions appear below: the *perpendicular band* is the reference foreground $\mathcal{F}(G)$ for band-fidelity metrics, and the whole frame for the region-count CDE (which is already gated by the affected regions A_G, A_R); the *grazing band* is the ± 12 -pixel strip $\{|s| \leq 12\}$ about the line. The definitions below apply unchanged to either evaluation set: three voxel-axis planes or five arbitrary-normal planes.

Band fidelity. S_b is the SSIM between the method render and the reference over the reference foreground $\mathcal{F}(G)$, computed per perpendicular view and averaged over the perpendicular views of all planes in the evaluated set. We also compute band PSNR but omit it from the main tables, as the operators differ by hundredths of a decibel and leakage can offset a small brightness bias.

Difference-referenced cut error (CDE). For a camera we form the *removal maps* of the method render R_{full}, R_{clip} and the reference G_{full}, G_{clip} (same camera, clip on and off):

$$D_R = Y(R_{full}) - Y(R_{clip}), \quad D_G = Y(G_{full}) - Y(G_{clip}). \quad (S10)$$

Thresholding at $\theta = 0.04$ gives the binary affected regions $A_R = \{|D_R| > \theta\}$ and $A_G = \{|D_G| > \theta\}$, and

$$CDE = \frac{\sum_v |(A_G \setminus A_R) \cap \mathcal{B}| + \sum_v |(A_R \setminus A_G) \cap \mathcal{B}|}{\sum_v |A_G \cap \mathcal{B}|}, \quad (S11)$$

the region symmetric difference (under-removal plus over-removal) normalized by the reference region, with the pixel counts summed over all views v of the family and all planes in the evaluated set before a single ratio is taken. Differencing each render against its own unclipped companion cancels the reconstruction floor that a direct clipped-vs-reference comparison carries, and the set difference prevents under-removal (a hole the method leaves unfilled) from being paid for by over-removal (material it wrongly deletes). For CDE_p the band $\mathcal{B}$ is the whole perpendicular frame (the A_G, A_R gating already localizes the score to the affected face); for CDE_g it is the grazing ± 12 -pixel strip.

Leak. On grazing views, Leak is the fraction of the method's near-plane foreground energy that falls on the culled side where the reference is background:

$$Leak = \frac{\sum_{\mathbf{x} \in \mathcal{L}} \max_c R_c(\mathbf{x})}{\sum_{\mathbf{x} \in \mathcal{N}} \max_c R_c(\mathbf{x})}, \quad (S12)$$

where $\mathcal{N} = \mathcal{F}(R) \cap \{|s| \leq 60\}$ is the method's near-plane foreground and $\mathcal{L} = \mathcal{N} \cap \{s > 2\} \setminus \mathcal{F}(G)$ restricts it to just past the cut edge, on the culled side, and off the reference foreground (where the reference is empty). The numerator and denominator are summed over all grazing views of the

evaluated set before the ratio is taken (pooling avoids the instability of averaging per-view ratios on thin grazing bands). Exact half-space truncation introduces no wrong-side mass in the represented Gaussian model. Image-space Leak uses an independently rendered reference mask and therefore measures proxy reconstruction and renderer mismatch alongside the soft tails or overshooting culls introduced by approximate operators. We interpret Leak jointly with S_b and CDE: together they couple culled-side energy with retained-face fidelity and over-removal.

Camera-independent geometric cut error (CErr_{3D}).

This diagnostic is computed in closed form from the primitives. For plane $(\mathbf{n}, \tau)$ and primitive (μ, Σ, α) , let $t = (\tau - \mathbf{n}^\top \mu) / \sqrt{\mathbf{n}^\top \Sigma \mathbf{n}}$; the exact kept opacity mass is $e_i = \alpha_i \Phi(t_i)$. We define each operator’s misplaced mass directly. Ours has $\epsilon_i^{\text{Ours}} = 0$ by the analytic definition, making CErr_{3D} a camera-independent operator-geometry diagnostic; Section S2 independently validates the finite-precision CUDA path. For HC,

$$\epsilon_i^{\text{HC}} = |\alpha_i \mathbb{1}[t_i \geq 0] - e_i|. \quad (\text{S13})$$

This is removed kept mass when HC drops the primitive and retained culled mass when it keeps it. Because MM’s total opacity mass is e_i , its misplaced mass is the moment-matched Gaussian tail beyond the plane,

$$\epsilon_i^{\text{MM}} = e_i \Phi\left(-\frac{t_i + \lambda(t_i)}{\sqrt{v(t_i)}}\right). \quad (\text{S14})$$

For the voxel-axis diagnostic in Table S3, we pool the three planes and normalize by the exact kept mass:

$$\text{CErr}_{3D} = \frac{\sum_i \epsilon_i}{\sum_i e_i}. \quad (\text{S15})$$

Σ is the full rotated covariance with the persisted Mip-Splatting filter, and t uses the exact evaluation-plane normals and offsets.

Table S3 reports the diagnostic on the Ours-trained and HC-trained interiors. MM can place only a soft tail beyond the plane, whereas HC can both remove kept mass and retain culled mass. This controlled diagnostic compares render-time rules on the shared N-DGS interiors; ClipGS remains in the image-space system comparison. On the Ours-trained interior, MM and HC average 0.225×10^{-2} and 1.400×10^{-2} ; on the HC-trained interior, they average 0.213×10^{-2} and 1.317×10^{-2} . The ordering is therefore unchanged by the checkpoint.

Arbitrary-normal geometric error. On the five arbitrary-normal planes per volume, XClipGS retains zero CErr_{3D} by construction, whereas MM and HC average 0.174×10^{-2} and 1.083×10^{-2} , respectively. Thus the geometric ordering is not specific to voxel-axis planes.

Metric-parameter sensitivity. For the voxel-axis set, we recompute both image-space metrics from fixed renders over a parameter grid. For CDE_g , the affected-region threshold is $\theta \in \{0.02, 0.04, 0.06, 0.08\}$ and the grazing-band half-width is $b \in \{8, 12, 16, 24\}$ pixels, giving 16 combinations; CDE_p uses the four thresholds and the full face. For

Table S3: **Camera-independent cut error.** $\text{CErr}_{3D} \downarrow$ ($\times 10^{-2}$) on both interiors; Ours is **0.000** by construction.

Scene	Both	Ours-trained		HC-trained	
	Ours	MM	HC	MM	HC
abdomen	0.000	0.166	1.015	0.166	1.015
intestine	0.000	0.191	1.201	0.196	1.210
knee-joint	0.000	0.217	1.347	0.204	1.249
lower-limb	0.000	0.205	1.264	0.196	1.183
vascular	0.000	0.216	1.373	0.216	1.367
heart	0.000	0.149	0.931	0.140	0.862
nose (MRI)	0.000	0.227	1.393	0.224	1.379
hand (MRI)	0.000	0.426	2.675	0.361	2.274
avg	0.000	0.225	1.400	0.213	1.317

Leak, we take the Cartesian product of foreground threshold $\theta_{\text{fg}} \in \{0.02, 0.04, 0.08\}$, cut-edge margin $\delta \in \{1, 2, 4\}$ pixels, and near-plane half-width $w \in \{40, 60, 80\}$ pixels, giving 27 combinations. Each setting uses the same pooled-within-scene, mean-across-scenes aggregation as the main tables.

Table S4 reports eight-volume ranges and per-method win counts. Ours has the lowest face CDE at all four thresholds and the lowest Leak in all 27 settings. Edge CDE is the close comparison: Ours is best in 11 of 16 settings and MM in the other five; ClipGS and HC are never best. Thus the face-localization and leakage conclusions are stable, while the small default edge-CDE advantage over MM is parameter-dependent. The main-table row uses $(\theta, b, \theta_{\text{fg}}, \delta, w) = (0.04, 12, 0.04, 2, 60)$.

S6 External-View-Only Ablation

Table S5 gives the per-scan breakdown of the clip-aware-supervision ablation. The clip-aware model is our exact operator trained on the full view set; the external-view-only model is the identical configuration trained on the intact views alone, with every clipped training view removed. Both are scored on the same held-out split at the fixed 30,000-iteration checkpoint (so CA reproduces the main-paper “Ours” column), reported here as intact-, clipped-, and all-view PSNR alongside the cut-face metrics for both models. Clip-aware supervision raises clipped-view PSNR on all eight scans, improving the average by 8.26 dB ($23.76 \rightarrow 32.03$), and substantially improves band SSIM and CDE_p . EX averages 0.72 dB higher on intact views, while the full model’s gain concentrates on the newly exposed cross-section. Leak stays comparably low because both models share the exact operator; band SSIM and CDE capture the interior-fidelity gain from clipped supervision. The cut-face columns use the voxel-axis evaluation set.

Table S4: **Metric-parameter sensitivity.** Ranges of eight-volume means ($\times 10^{-2}$); “Wins” counts settings with the lowest mean.

Metric	Settings	Ours	ClipGS	MM	HC	Wins
CDE _g	16	17.47–29.57	21.73–50.55	15.58–36.06	19.24–43.64	Ours 11; MM 5
CDE _p	4	11.97–25.74	13.18–28.33	12.14–26.36	12.42–27.01	Ours 4
Leak	27	0.20–0.72	12.60–20.96	4.24–8.64	9.56–16.05	Ours 27

Table S5: **Clip-aware supervision ablation.** CA uses full supervision; EX uses intact views only. iP, cP, and aP are PSNR on intact, clipped, and all views; cut errors and Leak are $\times 10^{-2}$.

Scene	CA = Ours (full supervision)							EX = ablation (external views only)						
	iP $\uparrow$	cP $\uparrow$	aP $\uparrow$	S _b $\uparrow$	Leak $\downarrow$	CDE _g $\downarrow$	CDE _p $\downarrow$	iP $\uparrow$	cP $\uparrow$	aP $\uparrow$	S _b $\uparrow$	Leak $\downarrow$	CDE _g $\downarrow$	CDE _p $\downarrow$
abdomen	33.55	33.05	33.30	0.899	0.26	28.2	34.4	35.03	23.01	29.02	0.325	0.79	39.8	51.8
intestine	32.59	30.96	31.77	0.920	0.16	22.4	17.2	33.58	20.84	27.21	0.340	0.27	22.8	24.3
knee-joint	33.52	24.76	29.14	0.825	1.45	15.2	11.1	33.62	18.40	26.01	0.244	2.16	18.3	17.3
lower-limb	34.53	34.04	34.28	0.875	0.02	19.2	19.0	35.35	26.56	30.95	0.555	0.03	21.1	24.4
vascular	34.30	38.46	36.38	0.876	0.06	25.4	13.1	34.75	31.19	32.97	0.416	0.08	27.5	20.4
heart	31.04	27.37	29.20	0.877	0.41	21.5	12.2	31.47	18.58	25.03	0.263	0.39	22.8	16.8
nose (MRI)	38.21	33.63	35.92	0.720	0.09	25.3	15.2	39.03	25.74	32.39	0.198	0.10	28.4	20.7
hand (MRI)	43.02	33.95	38.48	0.886	0.71	24.6	20.9	43.72	25.77	34.74	0.425	0.57	25.3	28.8
avg	35.09	32.03	33.56	0.860	0.40	22.7	17.9	35.82	23.76	29.79	0.346	0.55	25.8	25.6

S7 Operator-Swap Controls and Per-Volume Results

All operators use the same voxel-axis planes and cameras as the main operator-swap comparison.

Control design. We assess operator/checkpoint coupling in three configurations. First, the trained-system comparison scores HC and MM on their own co-adapted interiors. Second, the fixed-interior comparison supplies every render-time rule, including RaRa, with the same clip-aware interior. Third, Table S7 repeats the swap on the HC-trained checkpoint. Ours retains its average edge-error and leakage lead across the co-adapted and two fixed-interior comparisons.

Per-volume results. The main paper reports average operator-swap results on two fixed interiors. Tables S6 and S7 provide the corresponding per-volume measurements. In each table, every operator renders the same primitives with matched cameras, clipping planes, and Mip-Splatting forward. Metrics and units follow the main paper: S_b $\uparrow$ is cut-face band SSIM, CDE_g/CDE_p $\downarrow$ are difference-referenced cut errors at the edge and over the exposed face, and Leak $\downarrow$ is culled-side leaked energy. Error and leakage values are $\times 10^{-2}$.

On the Ours-trained interior, HC has the largest edge error CDE_g on all eight volumes, and Ours leaks least on all eight; on the HC-trained interior the same ordering holds, establishing consistency across both fixed interiors.

S8 Supplementary Demonstrations

Two demonstrations accompany this paper on the project page.¹ Neither is required to assess the paper: every reported

number comes from the CUDA implementation and protocol described in Sections S2 and S3, and the main paper is self-contained. Both show behavior that is inherently temporal or interactive and therefore cannot be conveyed by a static figure.

Video demonstration. Two screen recordings of the same held-out volume (intestine, CT) render interactively while a clipping plane sweeps through the anatomy, one using our analytic operator and one using the ClipGS reimplementation. Both are captured live from the trained checkpoints and are unedited. They show the temporal behavior that the still figures cannot: with the analytic operator the exposed face stays sharp and no material appears on the culled side, whereas the baseline leaks past the plane and its boundary flickers as whole primitives are kept or dropped. The measured culled-side leakage on this volume differs by roughly two orders of magnitude (0.16 versus 12.68×10^{-2} ; Table S1). Frame rates visible in the recordings come from a shared GPU and are not the timing measurements of Section S3.

Interactive demonstration. A self-contained web page renders a Gaussian scene and switches among the three clip operators at render time, with a movable plane and a sweep animation, so that hard-cull popping, the moment-matched tail, and the analytic boundary can be compared directly. It runs offline with no build step. This page is an independent WebGL reimplementation of the three operators for illustration; it is *not* the CUDA rasterizer used for any reported result, and it corroborates only the qualitative operator behavior described in the half-space clipping operators section of the main paper. The bundled rendering engine is third-party and included verbatim so the page requires no network

¹<https://gaozhongpai.github.io/XClipGS/>

Table S6: Operator swap on the Ours-trained interior. Bold marks the full-precision best.

Scene	Ours (analytic)				RaRa				MM				HC			
	S _b ↑	CDE _g ↓	CDE _p ↓	Leak ↓	S _b ↑	CDE _g ↓	CDE _p ↓	Leak ↓	S _b ↑	CDE _g ↓	CDE _p ↓	Leak ↓	S _b ↑	CDE _g ↓	CDE _p ↓	Leak ↓
abdomen	.899	28.2	34.4	0.26	.901	28.0	34.3	1.31	.895	31.0	35.1	3.05	.892	39.1	34.6	11.06
intestine	.920	22.4	17.2	0.16	.920	22.6	17.3	5.65	.917	20.9	17.2	2.65	.907	26.1	17.5	9.37
knee-joint	.825	15.2	11.1	1.45	.823	15.0	11.1	6.00	.805	15.9	11.0	15.31	.798	18.5	11.4	26.88
lower-limb	.875	19.2	19.0	0.02	.872	19.2	19.6	7.95	.864	19.8	19.0	2.48	.862	27.8	19.3	10.69
vascular	.876	25.4	13.1	0.06	.857	27.8	14.3	6.68	.867	25.1	13.1	0.74	.864	28.5	13.2	3.70
heart	.877	21.5	12.2	0.41	.878	21.9	12.5	6.62	.859	23.6	12.4	13.39	.846	27.6	12.5	25.93
nose(MRI)	.720	25.3	15.2	0.09	.715	27.1	15.5	6.80	.698	28.2	15.3	4.41	.678	34.1	15.4	12.60
hand(MRI)	.886	24.6	20.9	0.71	.882	25.8	21.4	8.94	.877	25.1	21.2	4.75	.854	39.0	21.6	18.91
avg	.860	22.7	17.9	0.40	.856	23.4	18.3	6.24	.848	23.7	18.0	5.85	.838	30.1	18.2	14.89

Table S7: Operator swap on the HC-trained interior. Bold marks the full-precision best.

Scene	Ours (analytic)				RaRa				MM				HC			
	S _b ↑	CDE _g ↓	CDE _p ↓	Leak ↓	S _b ↑	CDE _g ↓	CDE _p ↓	Leak ↓	S _b ↑	CDE _g ↓	CDE _p ↓	Leak ↓	S _b ↑	CDE _g ↓	CDE _p ↓	Leak ↓
abdomen	.895	28.8	34.2	0.29	.896	28.7	34.2	1.28	.897	31.3	34.8	3.39	.895	38.9	34.7	10.50
intestine	.909	22.3	17.6	0.16	.909	22.6	17.7	4.82	.910	20.9	17.4	3.20	.902	25.6	17.7	8.28
knee-joint	.761	15.9	11.9	1.36	.772	15.0	12.0	4.89	.755	14.9	11.6	12.69	.734	19.6	11.6	22.44
lower-limb	.850	19.4	19.7	0.02	.845	19.3	20.3	6.05	.846	19.1	19.4	1.75	.847	26.0	19.6	8.08
vascular	.848	25.6	13.7	0.06	.829	27.6	14.8	5.15	.847	25.2	13.6	0.72	.847	27.9	13.6	3.79
heart	.839	22.5	12.5	0.36	.839	22.8	12.9	5.45	.844	25.2	12.4	12.37	.826	29.2	12.4	23.00
nose(MRI)	.672	25.5	15.7	0.09	.667	26.9	15.9	4.87	.673	27.8	15.5	3.76	.665	34.0	15.6	9.62
hand(MRI)	.851	24.7	22.7	0.66	.852	26.0	23.0	6.61	.853	23.4	22.5	3.09	.838	34.1	22.7	14.63
avg	.828	23.1	18.5	0.37	.826	23.6	18.8	4.89	.828	23.5	18.4	5.12	.819	29.4	18.5	12.54

access.